

Team Members

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Project Title

AI-Powered Phishing Website Detection System Using Machine Learning with Real-Time URL Analysis

Project Description

Phishing attacks are one of the most common cybersecurity threats where attackers create fake websites to steal sensitive information such as passwords, credit card details, and personal data. Traditional phishing detection methods rely on blacklists, which fail to detect newly created phishing websites.

This project aims to develop an intelligent phishing website detection system using Machine Learning that can automatically classify websites as phishing or legitimate based on various features such as URL structure, domain information, and webpage content. The system will use supervised machine learning algorithms like Random Forest, Support Vector Machine (SVM), Logistic Regression, and XGBoost to train on a dataset containing phishing and legitimate website features.

The project will include complete end-to-end development, including data preprocessing, model training, evaluation, and deployment using a web interface. Users will be able to input a URL, and the system will instantly predict whether the website is safe or phishing.

The final system can be deployed as a web application, desktop application, or browser extension, helping users detect phishing attempts in real time and improving cybersecurity awareness and protection.

Project Objectives

1. To provide real-time phishing detection results.
2. Use deep learning models for improved accuracy.
3. Create an API for phishing detection.
4. Build a full stack web app using React + Fast API.